ChatTFT: building decision-relevant analytical workflows

Recommendation

Start with a decision-centered investigation loop, supported by a small, model-selected library of analytical playbooks—not a deep hierarchy of TFT topics.

The organizing question should be:

> ***What uncertainty is preventing a useful answer, and which available observation would resolve it?***

That is different from asking which skill matches the nouns in the prompt, or which tool can retrieve related statistics.

My recommended starting architecture has one agent responsible for the investigation, a compact record of the player's decision and supporting evidence, and selectively loaded playbooks for recurring analytical problems. Those playbooks should specify **when they apply, what they establish, how results change the investigation, and when to stop**. A hierarchy may eventually help navigate a larger library, but it should not determine the investigation's shape.

The attached files support a more specific diagnosis than "the assistant needs to plan better": **planning guidance already exists, but it competes with ranking-oriented routing, mandatory query sequences, and report templates that are not consistently tied to the player's actual decision.** These are design risks visible in the supplied instructions—not proven explanations of production behavior, because the attachment bundle does not include execution traces or the skill-selector implementation. system system SKILL SKILL

This assessment combines the attached source, primary research, and historical player discussions. I have not executed the ChatTFT database. The workflows below are proposed investigations, not claims about the current meta.

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1. What the existing system already does—and where direction can break down

The foundation is stronger than the earlier report recognized

The statistical-investigation skill already asks for the decision, population, baseline, controls, metrics, and a validating follow-up. The interpretation skill already warns about selection, survivorship, holder binding, and the limitations of final-board data. The cohort tools expose explicit comparisons, placement histograms, overlap information, and eligible lobby-clustered confidence intervals. These are substantial analytical capabilities. SKILL SKILL cohort_tools

Consequently, simply adding "identify the decision, make a plan, and consider confounders" would largely repeat existing instructions. The more useful question is **what prevents those principles from controlling the next action**.

Four concrete sources of friction

Source-derived observation	My assessment of the resulting risk

The main prompt routes questions answerable with a ranking toward immediate finalization; the analyst is introduced primarily through ranking and lookup examples.	A question may be classified by its easiest statistical approximation rather than its intended decision. “Best item?” becomes a leaderboard before the assistant considers whether the player means a third slot, a slam, or a line-enabling choice.
The itemization skill mandates calls to all three cohort-delta tools.	Tool coverage becomes an obligation independent of information value. The agent may investigate unrelated units and traits because the workflow requires it.
The composition and unit experts prescribe extensive report content and, for units, a fixed number of selections.	A narrow player question can expand into a generic composition report. Conversely, a report template can demand claims the evidence does not justify.
Several prompts describe 500 or more boards as “solid.”	Sample volume can become a substitute for examining the comparator, selection process, and precision of the particular claim.

These observations come directly from the main, analyst, itemization, unit-expert, and comp-expert instructions. system system SKILL system system

An especially important example: a correct delta can answer the wrong question

In the supplied implementation, `rank_items(holder=...)` restricts the **returned item rows** to a holder, but its underlying board population remains the active scope. Its attached delta compares the item-holder combination against boards without that combination—not automatically against boards containing the same holder without that item. That follows from `rank_items`, `ranking_board_context`, and `entity_delta_page`. ranking_tools utils

Therefore, an item-holder row can be perfectly correct while failing to answer:

> *“Is this item preferable to the alternatives on this unit?”*

The latter requires a deliberately chosen comparison population.

This is not necessarily a tool bug. It demonstrates the missing layer: **guidance that translates a player’s question into the right comparison, rather than merely teaching the model to read the returned fields correctly.**

The larger evidence boundary

The supplied cohort filters cover units, items, traits, and final level. Grouping exposes additional final-board dimensions. They do not expose decision-time HP, component inventory, streak history, shop offers, transition paths, or combat matchups. The root prompt documents a separate rolldown calculator, but its implementation is not included. models cohort_tools system

That creates a fundamental distinction:

“Which achieved boards performed well?” is not the same question as “Which action should this player take from this position?”

A stronger workflow should bridge that distinction where defensible and expose it where not. It should neither silently substitute the first question nor refuse every strategic question merely because causal

certainty is unavailable.

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2. What makes a workflow valuable?

I would define a valuable workflow as one that improves the player's **choice, understanding, or confidence calibration**, relative to the time required to obtain the answer manually.

That includes several different kinds of value:

- **Selection:** identifying the relevant alternatives and eliminating infeasible ones.
- **Conditional insight:** showing when a recommendation changes.
- **Interpretation:** explaining why an apparently strong statistic is misleading or insufficient.
- **Compression:** delivering a reliable lookup faster than manual navigation.

More tool calls, more statistics, and a longer explanation are not themselves value.

Consider three questions involving the same item and unit:

> *"What are the most common items on Bard?"*
> *"Bard already has two items; which of these two should fill the third slot?"*
> *"Is getting this artifact enough reason to commit to Bard?"*

They share entities but differ in their **decision structure**. The first needs frequency retrieval. The second needs constrained alternatives. The third needs an assessment of dependency, feasibility, and the distinction between successful end states and committing to a line.

Conversely, "Does this composition require its five-cost?", "Is this artifact essential?", and "Can I drop this trait breakpoint?" share a reusable analytical pattern: **assess whether an apparently important feature is necessary, replaceable, or merely associated with stronger finished boards**.

That pattern is a better candidate for reusable guidance than a broad "unit analysis" category.

Ambiguity should be resolved economically

I recommend three responses to ambiguity:

Proceed under a stated assumption when a conventional interpretation is adequate. "Best items on Bard?" can usually begin as a general itemization overview without a questionnaire.

Branch the answer when two interpretations are both useful and inexpensive. "Best" might reasonably mean placement performance versus popularity.

Ask a question when the missing detail materially changes the feasible options or the calculation. A rolldown probability without a budget is a different case from a general item overview without the player's HP.

The criterion is not whether context is missing. Context is almost always missing. It is whether obtaining it would materially improve the answer.

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3. What relevant research supports—and does not support

The literature supports useful ingredients, but does not establish a complete TFT architecture.

Research	Relevant finding or design principle	Implication for ChatTFT
ReAct	Interleaving reasoning and action lets observations inform subsequent plans; the paper evaluates this on question-answering and interactive tasks.	Plans should be revisable after evidence arrives, not merely written before a fixed query sequence. arXiv
Hierarchical task-network planning / SHOP2	Methods decompose tasks under preconditions and can express partially ordered subtasks. This is richer than a topic taxonomy.	Borrow preconditions, dependencies, and alternative methods—not necessarily a full symbolic planner. IPC
SkillsBench, original paper version	Curated skills improved average performance by 16.2 percentage points, but some tasks worsened. Focused skill packages performed better than comprehensive ones in its analysis.	Skills are a testable intervention, not an automatic improvement. The reported skill-count relationship is not proof that every application should load exactly two or three skills. arXiv
Agent Workflow Memory	Reusable workflows induced from experience improved performance on web-navigation benchmarks.	Successful analytical episodes may provide better reusable material than an ontology designed entirely in advance. Transfer to TFT remains untested. arXiv
Rational metareasoning	Computation can be selected according to its expected contribution to the eventual decision, accounting for computational cost.	Stopping is part of the reasoning problem, not simply reaching a tool-call limit. People at EECS
Human–AI decision-making research	In Bansal and colleagues' experiments, explanations increased acceptance of recommendations without increasing complementary performance, including acceptance of incorrect recommendations.	Measure whether players make better judgments—not merely whether answers sound insightful or inspire confidence. arXiv

Anthropic's skills design also separates lightweight discovery metadata from detailed instructions and supporting references. That provides progressive disclosure without requiring every task to descend through a deep tree. Anthropic

The synthesis is: **procedural knowledge, adaptive execution, and selective context loading are promising; "hierarchy" alone is not the mechanism that makes analysis useful.**

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4. Compare genuinely different approaches

The following assessments are my expected tradeoffs for ChatTFT, not benchmark results.

Approach	Expected player value and flexibility	Reliability	Implementation and maintenance	Latency
Improved single-agent instructions, no skill library	Potentially substantial gains from fixing contradictory directives; highly flexible.	Depends on the model consistently applying a compact analytical policy.	Lowest initial complexity; can become unwieldy as instructions accumulate.	Lowest orchestration overhead.
Fixed workflow templates selected by the model	Strong for recurring, well-defined tasks; weaker for unusual combinations.	Predictable within covered cases; risks executing unnecessary steps.	Easy to test initially; branching templates become expensive to maintain.	Predictable but can waste calls.
Hierarchical skill taxonomy	Helps navigation through a large catalog; cross-cutting questions may fit awkwardly.	Wrong early branch can hide useful guidance; inherited instructions may conflict.	Moderate implementation; taxonomy and duplicated content need maintenance.	Additional selection/loading steps.
Retrieval of expert worked examples	Captures question-specific judgment and stopping behavior with relatively little abstraction.	Sensitive to retrieving genuinely analogous cases rather than merely matching entities.	Moderate setup; examples need validation and patch-specific conclusions must be separated.	Retrieval overhead, potentially fewer exploratory calls.
Decision/evidence-driven playbook composition	Strong fit for mixed questions: select guidance according to unresolved analytical needs.	Explicit evidence requirements make failures easier to diagnose; model judgment still matters.	Moderate if implemented as small structured state, high if overbuilt into a large graph system.	Adaptive; can preserve quick paths.

Specialist multi-agent team	Potentially useful for broad, separable research or independent critique.	Risks duplicated queries, inconsistent populations, and information loss between agents.	Highest coordination and testing burden.	Usually more overhead; parallelism helps only for independent work.
Automatically learned workflow or skill library	Could adapt to recurring real usage instead of an anticipated taxonomy.	Depends heavily on feedback quality and protection against reinforcing bad analyses.	High initial evaluation and curation requirements.	Potentially efficient later; substantial offline development cost.

Where hierarchy genuinely helps

A hierarchy is useful when the selection problem is **finding relevant guidance among many modules**. It is less useful when the problem is **deciding what evidence would matter**.

Even an excellent tree such as `items → artifacts → holder analysis` does not establish whether the user needs an allocation lookup, an interaction investigation, or advice about committing to a line.

There are also three distinct hierarchies that should not be conflated:

A browsing hierarchy organizes documents.
A task hierarchy decomposes a goal.
An execution dependency structure determines what must be learned before another operation makes sense.

ChatTFT could benefit from the latter two while retaining a mostly flat skill catalog.

Why I would not start with more specialists or autonomous skill generation

The attached system already has several specialist prompts. I would first test whether one investigator with a shared decision record improves coherence before adding more handoffs. That is also consistent with engineering guidance to add agentic complexity only when simpler arrangements are insufficient. system system system Anthropic

Automatic workflow learning is worth revisiting after good evaluations exist. The 2026 SkillLearnBench preprint reports benefits from continual learning, but uneven results across tasks and drift under self-feedback alone. It does not justify allowing an unevaluated agent to turn its own successful-looking answers into authoritative procedures. arXiv

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5. Recommended structure: decisions, playbooks, methods, and facts

A. Keep a small investigation record

Before a substantial investigation, the model should establish a compact record like this:

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request:
  decision: Choose a third item from two available options.
  objective: Compare completed-build placement performance.
  constraints:
    - Named holder and star level.
    - Two already-fixed items.
    - Named composition shell, if supplied.

evidence:
  required:
    - Comparable completed-build outcomes.
    - Samples for both alternatives.
  limitations:
    - Final boards do not reveal acquisition timing.
  unresolved:
    - Whether the apparent ranking survives the specified build context.

next_step:
  question: Which option performs better within that context?
  operation: Retrieve constrained loadouts.
  branch:
    clear_supported_difference: Summarize the comparison.
    thin_or_mixed_evidence: Test one relevant relaxation or report uncertainty.
```

This is a proposed internal working structure, not a mandatory user-facing planning monologue. It need not become a new database or orchestration framework.

Its purpose is to make the connection between **question, evidence, and next action** inspectable.

B. Organize playbooks by analytical purpose

A starting catalog could include:

Playbook	Reusable problem it addresses
Compare feasible alternatives	Choose among named items, holders, board variants, or achievable end states.
Assess dependence and replaceability	Determine whether a feature is essential, optional, or a marker of stronger boards.
Investigate a surprising statistic	Examine whether a headline survives a meaningful change in population or grain.
Map a composition’s structure	Separate common identity, item allocation, variants, and selected high-cap versions.
Explore an unfamiliar opportunity	Generate a bounded shortlist, then validate the most consequential candidates.

Assess a sequential decision	Separate calculable probabilities, observable outcomes, missing transition information, and conditional judgment.

A straightforward lookup should need **no substantial playbook**.

These categories are proposals to test against cases, not a definitive ontology.

C. Keep methods separate from playbooks

Methods are smaller reusable operations: defining a composition shell, constructing a disjoint comparator, interpreting a placement distribution, checking holder binding, examining a suspected star-level mixture, or responding to suppressed results.

A dependency assessment might use several of these. So might a constrained item comparison. Neither should inherit an entire “statistics course” whenever one method is needed.

The model should select a primary playbook and then load methods only for identifiable evidence needs. Cross-links and dependency metadata are sufficient initially; a graph database is unnecessary.

D. Keep current game facts separate from workflow instructions

Item effects, trait definitions, current set membership, and patch-specific mechanics belong in versioned factual context. Procedures such as “compare alternatives within the fixed build” should not contain long-lived assumptions about particular units or items.

This is especially relevant because the attached itemization skill mixes investigation instructions with substantial mechanical and archetype descriptions. Those serve different maintenance and retrieval purposes. SKILL

E. Give every playbook a completion contract

A useful playbook should answer:

When does this apply? What evidence is required? Which claims can that evidence support? What observations change the next step? What counts as enough? When should the workflow exit without a recommendation?

For dependency assessment, completion might mean:

> *The analysis distinguishes “common,” “associated with better outcomes,” and “necessary for the line,” and either supports the strongest applicable claim or explains why necessity cannot be established.*

That is more useful than completion meaning “called all three delta tools.”

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6. Stress-test the structure against varied player questions

Historical player discussions show why entity-based categories are insufficient. One thread combines questions about rolling, preserving economy, and reaching level nine with limited HP. Another discusses a named trait package spanning multiple compositions, alternative damage directions, late-game caps, and mechanical interactions. These are examples of question diversity, not evidence that the commenters’ advice is correct. Reddit

The following is a proposed stress-test set. Capability judgments refer to the supplied tool interfaces, not every capability that may exist elsewhere in the repository. cohort_tools models ranking_tools

Representative question	Productive investigation	Appropriate boundary or stopping point
“What are the most-played four-costs?”	Frequency ranking with the requested cost filter.	Stop after the lookup; do not turn popularity into strength.
“Which four-costs have the best placement?”	Performance ranking with samples and comparable star grain.	Report near-ties or thin leaders without inventing strategic explanations.
“What does this artifact actually do?”	Retrieve verified static context.	Statistical queries cannot establish mechanics.
“Best items on Bard?”	General holder overview; clarify frequency versus outcome only as needed.	Avoid a full composition investigation unless a meaningful pattern demands it.
“I have two fixed items. Which third?”	Compare constrained completed loadouts.	Stop when the available alternatives are adequately distinguished—or not distinguishable.
“Tank or damage items on Shen?”	Examine role-consistent allocations, star profiles, and relevant board contexts.	Do not infer unit mechanics from item names alone.
“Does Vi need this artifact?”	Within-shell dependency assessment with holder-bound conditions.	Distinguish observed success without it from causal necessity.
“I hit this artifact at 2-1. Should I force the line?”	Assess compatible end states, current feasibility, and transition uncertainty.	Final-board outcomes cannot establish the optimal commitment policy.
“Which of these two holders should get my spare item?”	Compare feasible holders in relevant contexts; consider what is displaced.	Global holder rankings may not represent this allocation choice.
“Can this comp work without its five-cost?”	Separate the base shell from high-cap variants.	Do not define the shell using the unit whose necessity is being tested.
“Which two units can I replace?”	Examine actual alternative packages and preserved constraints.	Independent unit deltas do not establish the value of a simultaneous swap.
“Six trait or four trait plus a legendary?”	Compare operationally defined, comparable board variants.	Preserve the opportunity cost in board slots; avoid vague trait-versus-unit rankings.
“Is this a top-four comp or a first-or-eighth comp?”	Retrieve placement distributions.	A mean alone cannot establish either pattern.
“What looks underrated?”	Bounded exploration of performance, frequency, and	Treat discovery as provisional; do not select the most extreme row

	plausible explanations.	and declare a hidden meta.
“Roll now or push nine?”	Separate hit probabilities from survival and future-board value.	Missing temporal evidence prevents a reliable numerical policy value.
“Does being contested make this line bad?”	Distinguish known pool depletion from outcomes under contestation.	The supplied final-board filters do not directly observe decision-time contesting.
“What counters this lobby?”	Use supplied boards and verified mechanics where available.	Final placement aggregates are not head-to-head combat evidence.
“Which augment should I take?”	Use supplied mechanics and state; identify missing outcome evidence.	The attached cohort DSL has no augment predicates.
“Did the patch make this unit stronger?”	Requires aligned historical scopes and patch context.	The supplied fixed-scope tools cannot establish a cross-patch comparison.
“Why did I go sixth with perfect items?”	Separate observable final-board issues from possible path or positioning mistakes.	A final board cannot reconstruct the game’s decisive decisions.

The structural lesson is not that every question needs a new skill. It is that a small number of analytical patterns recur across many topics, while **answerability and temporal context change the permissible conclusion**.

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7. How the recommended approach would work

Example A: “Bard already has two items. Which third item?”

What matters: The player is choosing a marginal addition to a constrained build, not requesting Bard’s globally strongest individual items.

Resolve the named entities together. Use `rank_unit_loadouts` with the holder, fixed items, completed-item count, and relevant star level. A supplied trait constraint can also be represented. The tool supports these conditions directly. ranking_tools

The next step depends on the result. A well-supported comparison among the feasible options may suffice. A tiny leading sample or mixed build contexts may justify a targeted `compare_cohorts` call. An unavailable exact comparison may justify a clearly disclosed relaxation, but not silently dropping the player’s fixed items.

What not to do: Automatically retrieve unit and trait deltas, or combine the top three individual-item rows into a supposed best build.

Player-facing answer shape:

> *“For the build you already have, A is the better-supported third-slot choice. B’s stronger headline comes from a different build population. In your specified context, the relevant comparison is …”*

That sentence is conditional on obtaining such evidence; it is not a current finding.

Stopping: Stop when the feasible comparison is resolved, the evidence supports no clear distinction, or the remaining uncertainty cannot be addressed with available data.

Example B: “Does Vi need this artifact, or is it just her best item?”

What matters: “Best-performing,” “commonly built,” and “necessary for viability” are different claims.

Define the intended Vi shell independently of the artifact. Compare holder-bound artifact presence against an explicit alternative or complement within that shell. Check whether different star levels or materially different allocations are mixing distinct versions of the unit.

Only investigate traits or teammates when there is a specific reason to suspect that the artifact is concentrated in a different line. The delta tools can identify candidate differences; they should not automatically convert those differences into explanations. Their outputs are frequency-ordered, and missing comparators produce unavailable rather than neutral deltas. deltas

Useful possible conclusions: The ordinary version has substantial successful samples; the artifact version appears to be a distinct line; or the available comparisons are too selected to establish dependence.

Stopping: Once those interpretations are adequately distinguished, a broad sweep of other entities adds little.

Example C: “Can I play this composition without its five-cost?”

What matters: Whether the five-cost defines the composition or merely appears in its strongest finished versions.

Construct a shared shell that excludes the disputed unit. Compare boards with and without it, then inspect the most relevant alternative structures. Final-level cuts may help describe different achieved-board versions, but should not be described as fully controlling for how players reached them.

There is an important distinction between:

> *“The shell has successful observed finishes without the five-cost.”*

and:

> *“You should avoid rolling for the five-cost from your current position.”*

The first can be supported by final-board evidence. The second requires a prospective decision analysis.

The attached composition prompt already recognizes identity versus ceiling; the improvement is making that distinction control the investigation rather than making it one section of every composition report. system

Example D: “Roll now or push nine? I have 32 HP and 42 gold.”

What matters: This is a sequential choice involving survival, attainable upgrades, and future resources—not a comparison between final level-eight and level-nine placement averages.

A productive workflow identifies the actual upgrades sought and the resources left under each option. The documented rolldown calculator can support an appropriate probability subproblem, subject to its

inputs and exclusions. Its presence in the root prompt does not establish a model for survival, combat strength, or the value of leveling. system

The assistant can still contribute: calculate the supported probability, identify which missing observation matters most, and state a conditional decision rule grounded in verified mechanics or supplied state.

It should not multiply a hit probability by observed completed-board placement averages and present the result as the expected placement of rolling. Those quantities describe differently selected populations.

Stopping: Stop when the remaining uncertainty is about unobserved combat or transition dynamics, rather than generating additional final-board statistics that cannot resolve it.

Example E: “Find something underrated in this composition.”

What matters: This is exploratory. The assistant must generate candidates, but it also needs protection against mistaking search artifacts for discoveries.

Define the shell, inspect a bounded candidate set, and select promising observations using both frequency and outcomes. Then examine the most consequential alternative explanation: different star mix, itemized versus unitemized usage, or concentration in a rare capped version.

A second view of the same data is a **sensitivity check**, not independent confirmation. Adaptive analysis can overfit through repeated hypothesis selection, even when individual estimates look reasonable. arXiv

A useful result may be:

> *“This is a plausible variant worth testing, not an established upgrade. It remains interesting after the relevant comparison, but the present evidence cannot rule out selected high-roll boards.”*

Stopping: Return a small, defensible shortlist once the leading candidates have survived meaningful checks. Do not keep searching until something produces an impressive delta.

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8. Interpretation and stopping need stronger instructions

Do not treat “more controls” as automatically better

The existing interpretation skill asks whether a finding holds after controlling for many variables, including placement, final board state, and economy. That wording needs careful qualification. SKILL

Causal research shows that adjustment can remove bias, introduce bias, or block part of the effect being studied. Variables affected by the decision are not interchangeable with pre-decision confounders. UCLA FTP

For TFT, my application of that principle is:

> ***Use final-level and final-board cuts to describe comparable achieved states. Do not imply that these cuts recover the causal value of an earlier decision. Never control for the outcome merely to “make the comparison fair.”***

Likewise, reportability and confidence should remain separate. The tool’s 50-board floor is a reporting restriction, while the prompts’ larger-sample labels are heuristics. Neither establishes that an observational comparison is appropriate for the decision. utils system

Use a practical value-of-information test

Before a nontrivial follow-up, ask:

> ***What plausible result would change the answer, change its confidence, or reveal an important limitation?***

If no plausible result would do any of those, the call probably does not belong in the investigation.

Conceptually:

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[\text{Value of next query}

\mathbb{E}[\text{improvement in the supported decision or judgment}]

\text{latency, cost, and attention burden}.] $

This adapts the metareasoning principle; it is not a proposal to ask an LLM to invent numerical expected values. People at EECS

I would encode four legitimate stopping outcomes:

Resolved: The decisive question has sufficient support.
No clear distinction: The data do not establish a meaningful preference.
Evidence ceiling: The remaining uncertainty requires unavailable observations.
Budget reached: Further useful investigation exists, but exceeds the appropriate budget; return the supported partial result.

“No clear distinction” must not be confused with proof of equivalence. “Evidence ceiling” must not be disguised as a confident recommendation.

A compact core instruction worth testing

> *Identify the player’s requested judgment and the feasible alternatives. Use the cheapest evidence that can resolve it. Before extending an investigation, identify the unresolved question and how a plausible result would change the answer. Preserve the user’s constraints and distinguish observed outcomes, mechanism-based reasoning, and unsupported causal claims. Load additional guidance only for a concrete analytical need. Stop when the decision is adequately supported, the alternatives remain unresolved, or available evidence cannot answer the remaining question.*

This should replace contradictory instructions, not merely be appended to them.

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9. Evaluate player value, not procedural compliance

Start with diagnostic comparisons

I would initially compare three versions using the same model, tools, and frozen data:

Current system.
Simplified core instructions without new skills.

Simplified core plus a small model-selected playbook library.

Only then compare a flat catalog against hierarchical navigation or example retrieval, keeping the underlying guidance as comparable as possible. Otherwise a better-written library can make an inferior selector appear successful.

Add an **expert-selection condition** for a subset of cases: supply the appropriate playbook directly.

That creates useful diagnostic distinctions:

Result	Likely development implication
Expert-selected guidance does not help	Improve the guidance or reconsider the intervention.
Expert selection helps but model selection does not	Selection or discoverability is the bottleneck.
The plan is appropriate but the query represents the wrong population	Improve tool semantics and query construction.
Evidence is correct but the final answer overclaims or omits the choice	Improve synthesis and handoff preservation.
Extra analysis rarely changes the useful conclusion	Simplify or tighten stopping behavior.

These are experimental interpretations to investigate, not automatic causal diagnoses.

Build cases around decisions and failure modes

A practical initial suite could contain roughly 60 cases spanning quick lookups, constrained choices, dependence questions, open exploration, and unsupported or ambiguous requests. Begin smaller with real failures, then expand coverage.

Freeze the data snapshot and supply evidence-backed grading notes. Include cases where the strongest-looking row is misleading, a comparator is suppressed, cohorts overlap, or the correct response is a useful partial answer.

Do not require one exact tool sequence. Grade necessary evidence and semantic constraints. Anthropic's evaluation guidance explicitly warns that rigid sequence-based grading can penalize valid alternative solutions and recommends combining deterministic checks with calibrated judgment. Anthropic

Score the dimensions that matter

Dimension	Example evaluation question
Decision alignment	Did the answer address the player's actual choice rather than an easier neighboring question?
Evidence validity	Were scope, holder binding, grain, denominators, and comparators correct?
Incremental usefulness	Did it add a consequential distinction beyond the obvious lookup?

Calibration	Did certainty and causal language match what the evidence supports?
Efficiency	Did additional calls and reading time produce useful information?
Robustness	Does the behavior remain appropriate across repeated runs and equivalent phrasings?

Fabricated measurements, unsupported causal claims, and silently changed constraints should be serious failures regardless of presentation quality.

Repeated trials matter. The original τ-bench explicitly evaluates consistency across attempts, rather than treating one successful run as sufficient evidence of reliability. arXiv

Measure the stopping policy directly

For multi-step cases, preserve intermediate evidence states. Have reviewers compare an answer possible at an earlier state with the final answer.

Ask: Did the additional investigation discover a relevant reversal, reduce an important uncertainty, or merely add supporting-looking numbers?

This gives a concrete measure of whether deeper analysis earns its cost.

Finally, test against manual lookup

A player study should compare ChatTFT with a manual statistics workflow under comparable conditions. Measure task completion time, correctness of the supported conclusion, recognition of evidence limits, and unnecessary confidence—not just preference for the prose.

For ambiguous decisions without an objective best action, evaluate whether participants understand the tradeoffs and uncertainty. Do not fabricate a gold-standard recommendation.

This matters because more persuasive explanations can increase reliance without improving decisions. The human–AI research discussed earlier makes that a real evaluation concern, not merely a stylistic preference. arXiv

I would not use ladder performance as the first success metric. First establish that the system improves the immediate analytical task.

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10. Practical starting point and assumptions to test

First: remove the clearest sources of misdirection

Replace the mandatory three-delta sweep with result-dependent follow-ups. Make quick finalization depend on whether the lookup answers the **intended question**, not whether related numbers can be retrieved. Make composition-report sections conditional on relevance and support. Preserve canonical name resolutions during an investigation rather than repeatedly resolving unchanged names.

Keep the existing structured result contracts, suppression handling, and explicit comparison semantics. They are useful foundations, even if the higher-level workflow design changes. SKILL system system models

Second: prototype only a few high-value playbooks

I would begin with **constrained item/build comparison**, **dependence and replaceability**, and **investigation of surprising statistics**.

These span several entity types, expose important baseline errors, and can be exercised substantially with the attached tools. Include a direct-lookup path and an evidence-ceiling path alongside them.

Each playbook should have expert-reviewed examples showing not just a successful deep investigation, but also a case that stops early and a case that cannot support the desired conclusion.

Third: let observed failures determine the next structural change

Test these assumptions before expanding:

Decision recognition: Can the model distinguish a lookup, a constrained choice, and a prospective strategy question when they mention the same entities?

Evidence usefulness: Do the available final-board observations support enough consequential distinctions to improve player value?

Guidance selection: Can the model select playbooks by analytical need rather than topical similarity?

Composition: Do combined playbooks remain coherent without repeating instructions or conflicting over scope?

Stopping: Can the model recognize when another query would be useful—and when more statistics would only obscure the evidence ceiling?

Human value: Do players reach better-supported conclusions faster, rather than simply rating the answers as more sophisticated?

Bottom line

The strongest opportunity is **not a larger catalog of TFT instructions**. It is a tighter relationship between the player's question, the comparison that matters, the evidence available, and the next investigative step.

A skill hierarchy may become useful infrastructure. It is not yet the most important hypothesis to validate.

First establish that a small set of well-defined, selectively applied analytical procedures improves decisions over both a cleaned-up single-agent baseline and manual lookup. Then grow the organizing structure around demonstrated reusable workflows—not around the game's nouns.